

IDENTIFICATION

CODE : GI-4-S1-EC-QLT
ECTS : 1

HOURS

Cours : 0h
TD : 12h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 14h
Travail personnel : 6h
Total : 20h

ASSESSMENT METHOD

1 two-hour exam

TEACHING AIDS

Handout in English, exercises in English, Minitab software and an online tutorial in French

TEACHING LANGUAGE

French
English

CONTACT

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AIMS

GI-4-S1-UE-AMCI
GI-4-S1-EC-QLT
Quality - Maintenance

Understand the issues and tools of quality (quality process vs. quality tools) and maintenance (impact of the organisation of quality on the other functions of companies).

Understand the usefulness and the operation of some tools (control charts, Gage R&R, reliability, etc.).

This EC contributes to:

- * School skills in Engineering Sciences / Level (1 for sensitization, 2 for appropriation, 3 for mastery) :
 - . Solve questions asked by a scientific problem by being aware of limits / 3
 - . Develop an experimental approach / 3
 - . Implement a scientific approach / 2
- * School skills specific to GI:
 - . Observing, measuring, analyzing and interpreting an activity or a system from data / 3
 - . Managing a production system and react to malfunctions / 3
 - . Developing and implementing an action plan as part of a quality and continuous improvement approach / 3
 - . Collectively managing a project: organization, communication, group coordination / 1

CONTENT

FR or EN - possibility to integrate a group with lecture/practical work in english (but online tutorial in French only)

ONLINE TUTORIAL about Six Sigma to do without the assistance of the teacher
QUALITY

- * General introduction: relations of quality and maintenance
- * Introduction to quality: definitions, goals, voice of the client, overview of tools
- * Six sigma : Introduction, Etapes du DMAIC, Organisation.
- * Overview of ISO9001:2008
- * Control charts
- * Gage R&R

MAINTENANCE

- * Introduction: Definitions, Types of maintenance
- * TPM (Total Productive Maintenance), Overview, The 8 pillars with details on Autonomous maintenance in Pillar 2 and overview of RCM (Reliability-Centered Maintenance) in Pillar 3 (Planned maintenance), Lean Six Sigma
- * Types of maintenance, Impact of OEE (Overall Equipment Effectiveness), Frequency of change calculated from reliability function

BIBLIOGRAPHY

Bufferne J. (www.jean-bufferne.com), Le guide de la TPM, Éditions d'Organisation, Eyrolles, 2006.

Duret D., Pillet M., Qualité en production; De l'ISO 9000 à Six Sigma, 3e édition, Editions d'Organisation, 2005.

Pillet M., Six Sigma; Comment l'appliquer, Editions d'Organisation, 2004.

PRE-REQUISITES

Probability - Statistics